

MODEL QUESTION PAPER**Power Plant Engineering****Time: 3 Hours****Max Marks: 75****I. Answer all the following questions****(9 x 1 = 9 Marks)**

1	Write the names of any two thermal powerplants in India	M 1.02	R
2	Steam turbines are used to convertenergy intoenergy	M 1.03	U
3	What is the function of a Surge tank in a hydroelectric power plant.	M 2.01	R
4	Mention any two advantages of a diesel power plant	M2.03	R
5	Expand the term FBR.	M 3.06	R
6	Write the fission Reaction of U-235.	M 3.02	U
7	Name the act in India for boiler operation.	M 4.06	R
8	Mention any two pollutants in air due to combustion of fossil fuels.	M 4.01	R
9	List various greenhouse gases.	M 4.02	R

II. Answer any Eight questions from the following**(8 x 3= 24 Marks)**

1	Define Octane number and Cetane number. Mention their significance.	M 1.04	U
2	Explain the process of Regeneration in Thermal power plant.	M 1.03	U
3	Mention the Properties of a good fuel.	M 1.04	R
4	Compare HCV and LCV.	M 1.04	U
5	What are the advantages of combined cycle power generation.	M 2.05	R
6	Explain a fission chain with an example.	M 3.02	U
7	List the social and economic issues of Thermal power plants.	M 4.02	R
8	How acid rain is formed? What are its harmful effects?	M4.02	U
9	Comment on thermal pollution from power plant.	M 4.03	U
10	Describe the safety measures in a Chemical Handling system.	M 4.05	R

III. Answer all questions from the following (6x 7 = 42 Marks)

1	Explain the working of thermal power plant with line diagram.	M 1.03	U
	OR		
2	Explain the working of a Bomb Calorimeter with figure.	M 1.05	U
3	Explain the working of a hydroelectric power plant with general layout.	M 2.01	U
	OR		
4	Explain the working of a closed type Gas turbine power plant with neat figure.	M 2.05	U
5	Describe the applications of diesel power plant.	M 2.03	R
	OR		
6	What are the advantages and disadvantages of a diesel power plant?	M 2.03	R
7	Explain the main parts of a nuclear reactor with figure.	M 3.04	U
	OR		
8	List various coolants and their essential qualities for a nuclear power plant.	M 3.05	U
9	Explain the characteristic features of a PWR.	M 3.06	U
	OR		
10	Explain FBR with neat figure.	M 3.06	U
11	Explain Greenhouse effect.	M 4.02	R
	OR		
12	Describe the safety policies to be observed in power plants.	M 4.04	R